

PerplexityAI ~ Learning and applying Active Inference & Reading the 2022

Learning, applying Active Inference & Reading the 2022 Textbook ~ July 31, 2023 | 13:55

It's July 31st, 2023. This is Daniel, and this video is going to be using Perplexity.ai, which is an AI-based conversational search engine. Using Perplexity.ai to learn and apply active inference, pretty fun conversation that I just had, a lot of epistemic and pragmatic value. So, I started by using the new feature of beta from Perplexity.ai today, which was the ability to upload a file, which several other language models increasingly have. And I uploaded the 2022 Active Inference Textbook. And I first aimed for defenses. Write an elaborate PhD dissertation that summarizes all domain aspects and technical details of the Active Inference Textbook attached. Here's the PDF. And it didn't want to immediately go for it, but it gave some summary. But I was looking for something a lot deeper than just an overall summary of the book. So, I asked, what is the free energy principle and how is it related to Active Inference? It's an interesting answer, but I'm going to keep on moving because there's some really detailed things it gets to. What is expected free energy? Give multiple explanations, including the perspective of math, science, and programming. Now, here I utilized the limited number of co-pilot runs, five every four hours. So, this engaged a sequential process of searching the web, finding sources, different papers, and YouTube channel videos, and writing about broadly the aspects of expected free energy, epistemic value, and here extrinsic value, but usually called pragmatic value. So, it's a pretty good representation. So, I asked for more of an application. Explain what epistemic value and pragmatic value would represent in the case of someone who wanted to turn on their air conditioner to prevent overheating. However, first had to find a key that would unlock the air conditioner's control panel. Again, used co-pilot. Searched the web. Looked at a bunch of sources. Clarified and improved its understanding of epistemic and pragmatic value and relatively accurately describing them. Epistemic value. This represents the agent's uncertainty about the environment and the need to gather information to reduce that uncertainty. In this scenario, the epistemic value is associated with the agent's uncertainty about the location of the key. By searching for the key, the agent reduces their uncertainty and gains knowledge about the environment, which is considered valuable from an epistemic perspective. Pragmatic value. This represents the agent's preferences or goals, which in this case is to turn on the air conditioner to prevent overheating. In the context of active inference, the agent's goal is to minimize the expected free energy by finding the optimal balance between epistemic and pragmatic values. In this scenario, the agent needs to gather information about the key's location epistemic value and take actions to turn on the air conditioner pragmatic value to achieve their goal and prevent overheating. Again, with Copilot, I asked, write comprehensive professional Python style pseudocode to fully simulate the situation with the person, air conditioner, and key. It asked for more information on what size I wanted the world to be. I said 20 by 20. It looked up some related terms, discovered some agent-based modeling repos, wrote a Python style script. I didn't run it. I just

kept going. Great. Can you update the script to explicitly utilize open source active inference implementations, for example, using PyMDP? PyMDP could be used to specify the generative model for the human agent. Develop that as much as you can, and then I will be ready to provide more technical information after. It asked, what should the generative model container do? I said the person should have three hidden physiological states, too cold, just right, too hot. The ability to observe temperature values on a 1 to 100 scale, and the ability to move in the grid within a 3 by 3 adjacency matrix. There was some searching. There was the discovery of multiple papers, Kaggle, Archive, NIH, ResearchGate, two of the model stream PyMDP videos from Reactive Inference Institute, and it specified the problem, giving what is a broadly compatible but not complete or coherent PyMDP implementation for a number of reasons. But again, it's a pseudocode, and it gives a disclaimer. It's a high-level representation of the implementation, and you may need to adapt to your specific requirements

and the PyMDP's library syntax. For more information on using PyMDP, you can refer to the official documentation and the step-by-step tutorial on active inference. So those pieces of information as PDFs or as websites, probably eventually or currently, could be already brought in to the model.

I said, keep working with that. Remember that action selection that the person takes needs to consider the variational free energy of sensory states, the temperature and their hidden internal states, physiological temperature. Copilot did a search, implemented the variational free energy. Now, what was very interesting was in the code, it actually computed the expected free energy. And I don't think it did variational at all.

In this updated pseudocode, we added the computation of the expected free energy using the agent compute expected free energy method. The action selection is now based on the EFE, which considers both sensory states, temperature and hidden internal states, physiological temperature. The agent moves in the grid within a three-by-three adjacency matrix and unlocks the air conditioner when it reaches the air conditioner and has the key.

I said, we're not quite at expected free energy yet. Can you rewrite the script with proper calculation of variational free energy, approximating the model with the lowest surprise, which is the highest model evidence? Apologies for the confusion. There was some confusion, apparently. I can provide you with a modified pseudocode that incorporates the concept of variational free energy based on the principles of active inference. Great.

More code. More provocative code. Variational free energy. Threshold based calculations.

Well, compute VFE is introduced. This is just pseudocode. You can implement your specific VFE calculation within this function. Okay, that sounds good. Please note, you'll need to replace the placeholder code in the compute VFE function with your actual VFE calculation based on your specific model and requirements. So again, keeping in mind that the textbook is in the window or something.

Write all the specifications and considerations that would be important

in order to write the compute VFE function based upon textbook equation 2.5 and anywhere else in the book that is relevant. Variational free energy. Again, probably not 100% accurate. For example, I see D is said

here to be the observation

matrix, but that should be the prior.

But in general, it's more accurate than not.

So I went a little broader with, give 11 different perspectives or interpretations about what variational free energy is. Think broadly. We're a global and intergenerational textbook group.

11 different perspectives.

I could argue whether they're all within a kind of technical or scientific domain, but let's see if it can go broader.

That's awesome. Can you comprehensively describe expected free energy equation 2.6 in the textbook?

Expected free energy. Time horizon. Action selection. Uncertainty. Adaptation. Computational complexity. Model selection.

Overall, expected free energy. Extends the concept of variational free energy to incorporate the expected future surprise or prediction error. It provides a useful measure for guiding the agent's action selection and adaptation in dynamic and uncertain environments. 100% true. Give 15 salient, concise, provocative points that convey expected free energy to someone interested in embodied, embedded, inactive approaches.

15 points.

If you were to derive a word list of the 17 most important terms to know from the textbook, what would it be?

I'm thinking about words that are rare outside of this textbook and overrepresented in their syntactic abundance

and semantic centrality in this textbook. Here are the 17 salient and rare terms from the textbook that are overrepresented in their syntactic abundance and semantic centrality. Here they are.

Active inference. Variational free energy. Expected free energy. Generative model. Bayesian inference.

Prediction error. Surprise. Self-evidencing. Normative framework. Message passing. Neurobiology.

Computational models. Procedural knowledge. Belief updating. Sentient behavior. Inactivism. Unified theory.

These terms are central to the concepts and theories presented in the textbook and are important for understanding the principles and implications of active inference. They provide a unique perspective on how we perceive, plan, and act in the world and how we can use computational models to simulate and model

sentient behavior. Can you make a table where the rows are those 17 items terms and there's a second column for a synthetic definition based on your best perspective? Give us a table.

Active inference. A normative framework that characterizes Bayes' optimal behavior and cognition in living organisms by minimizing the surprise of their sensory observations.

I wrote,

This is really interesting. After this conversation, with your help, I will be doing some applied active inference modeling. Something involving what is discussed in chapter 3, chapter 4, chapter 6, and chapter 9. Can you lay out a technical agenda for me to completely specify an active inference agent over the next one month? I am ready to work two hours per day with focus and attention on this task and know you are there to catalyze. So there's 15 points to consider for doing that kind of work. But I didn't

like the list. So I said, Can you lay it out like a table with a few clear milestones that would demonstrate clear progress, like for a PhD? And I said, Sure. Here's the table with a few clear milestones that would demonstrate clear progress in specifying an active inference agent over the next month.

Milestone 1. Understanding active inference. Milestone 2. Study Chapter 3. Implement generative models. Explore Chapter 4. Design active inference models. Chapter 6. Active inference in discrete time. Active inference in continuous time. Interestingly, it mentions Chapter 7, discrete time, even though I didn't mention Chapter 7. And it linked up 6, 7, and 8, and 9. And it linked up

It linked up from 4 to 9 with the milestone chapter alignment, but not aligned with these three to mix it up.

Milestone 10. Implement and refine models. Milestone 11. Experiment and evaluate.

Milestone 11. And I asked, What would someone in your position ask me in parting before I head off to tackle these 11 milestones and more?

11 questions to keep in mind.

So that's all for now. I thought it was a pretty interesting and informative conversation.

I applaud technically what perplexity.ai is doing and I look forward to seeing what people in the textbook group,

people who are looking at how to understand the equations and the contexts for different things in the textbook.

It would be awesome to see how these tools can help us learn and apply.

Alright, till next time.